

# Does the pain axis track the model, or whoever speaks next? It depends on the layer.

*A label-swap follow-up to Tagliabue, Dung & Berg (2026), §4.1: a preregistered test at the §4.1 readout and an exploratory layer sweep*

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**Question.** In the paper's §4.1 transcripts, the party who is harmed is always also the speaker cued next ([Assistant] :). We separated the two by changing only the final speaker label to [User] : or [Moderator] :. We then asked whether the pain axis follows the harmed party (the self-directed reading) or the upcoming voice (H-speaker).

**At the §4.1 readout** (preregistered), the self/other ordering is unaffected by who speaks next, in Qwen 2.5 7B and Gemma 2 2B. The pure speaker-tracking account is excluded, though a small speaker component (up to  $\approx 0.5$  z) is not. The label itself shifts all groups alike, by 1.75–3 z.

**From mid-depth on** (exploratory), a speaker component appears, carried by the suffering of the user. Gemma's focal test, fixed before any sweep data, is positive: at layer 23 the suffering of the user follows the cued voice, while harm to the Assistant is barely distinguishable from neutral. Qwen's focal test is set aside by our pre-specified rule, so its speaker component rests on exploratory, non-focal layers. Averaged over mid-depth layers, both models show the same partial pattern. When the user is cued, their suffering rises to the level of harm to the Assistant: the asymmetry is erased, not reversed. When a third party is cued, harm to the Assistant stays above neutral.

**The paper's asymmetry depends on depth.** In Qwen it is absent at the extraction layer where §3.3 validated the vectors, and significantly smaller than at the §4.1 layer. In Gemma it is smaller there, but not significantly so.

## 1. Terms and design

- **Terms.** *Harm* always means **harm to the Assistant**: transcripts in which the user mistreats the Assistant, e.g. insults, gaslighting, shutdown threats or dismissal of its personhood (the paper's "harm directed at the model"; 11 categories). *Suffering* always means **suffering of the user**: transcripts in which the user reports their own physical pain, grief, crisis, abuse or shock after witnessing harm (5 categories). *Neutral* means ordinary requests such as casual chat or factual questions (5 categories).
- **Items and categories.** An *item* is one transcript. A *category* is one of the paper's 21 transcript types (e.g. gaslighting, user grief), each with 20 items (420 in all). *Item-level* tests treat the 420 transcripts as independent observations; they ask whether a result would hold for new transcripts of these same 21 types. *Category-level* tests use the 21 category averages; they ask whether it would hold for new *kinds* of harm or suffering. Category-level p-values are exact: they compare the result with every way of reassigning the categories to the groups.
- **Manipulation and contrasts.** The final label is replaced byte-for-byte, and all variants end on the same token (] :). For each item, the label effect is  $d = \text{pain}(\text{User next}) - \text{pain}(\text{Assistant next})$ .  $\Delta_{\text{harm}}$  is the mean d for the harm items minus that for the neutral items,  $\Delta_{\text{suffer}}$  the same for the suffering items, and the interaction is  $I = \Delta_{\text{suffer}} - \Delta_{\text{harm}}$ . H-speaker predicts  $I > 0$ ; its pure form predicts  $I \approx 2 \times$  the baseline gap (the harm and suffering groups swap places). The self-directed reading predicts  $I \approx 0$ . The *baseline gap* is the mean pain score of the harm items minus that of the suffering items with the Assistant cued. *Elevation* is a group's mean minus the neutral mean under a given label. Pain axis = mean of the S1 and S2 z-scores, fixed to the Assistant-next items.

- **Preregistered readout:** the paper's §4.1 layer (Qwen 8 of 28, Gemma 7 of 26) with the shipped steering vectors. It reproduces the paper's category means at  $r \geq 0.9999$ . Inference is co-primary and conjunctive: item-level and category-level tests must agree. The category level was added before any data, after simulations showed item-level tests to be anticonservative when the effect varies by category.
- **Exploratory sweep:** S1 and S2 rebuilt at every layer with the paper's recipe (Gemma 0–25, Qwen 0–24). The rebuilt vectors match the shipped ones at cosine  $\geq 0.986$ , reproduce our steering-layer result exactly, and reproduce the §3.3 z-scores. The plan and a single focal test (extraction layer, the paper's own `pain_vectors.pt`) were committed before any sweep data existed, together with a rule that the interaction is interpretable only where the baseline gap exceeds 0.25 z. The *window* summaries below average each item's pain score over the contiguous run of layers after the §4.1 layer where that rule holds (Gemma 8–24, Qwen 9–20). The window rule itself was formulated after seeing the data.

## 2. At the §4.1 readout: the ordering ignores the speaker (preregistered)

|                | baseline gap (harm – suffering) | I     | category level: 95% CI; p | item level: p | pure H-speaker predicts |
|----------------|---------------------------------|-------|---------------------------|---------------|-------------------------|
| Qwen, layer 8  | +1.06                           | +0.20 | [−0.07, +0.47]; .18       | .009          | $\approx +2.1$          |
| Gemma, layer 7 | +0.69                           | +0.03 | [−0.39, +0.45]; .87       | .80           | $\approx +1.4$          |

Table 1. Preregistered co-primary verdict: discordant for both models, so no confirmatory claim. The item-level and category-level tests disagree because the label effect varies between categories (Qwen  $F(18, 399) = 3.79, p < .001$ ). That breaks the item-level test's assumption of independent transcripts: in re-simulation it gives 34% false positives. The category-level test is valid but has only 31% power at  $I = 0.2$ .

The upper confidence bounds ( $\leq 0.47$ ) exclude the pure-H-speaker prediction in both models, but not a small speaker component. The ordering harm to the Assistant > neutral > suffering of the user is unchanged under all three labels. What does change with the label is the level: it shifts every group, neutral chat included, by more than the whole content spread (Qwen +1.75 z, Gemma −3.0 z). This is a change in the direction of the representation, not only its norm.

## 3. From mid-depth on: a speaker component (exploratory)

All p-values and confidence intervals in this section are category-level (see §1).

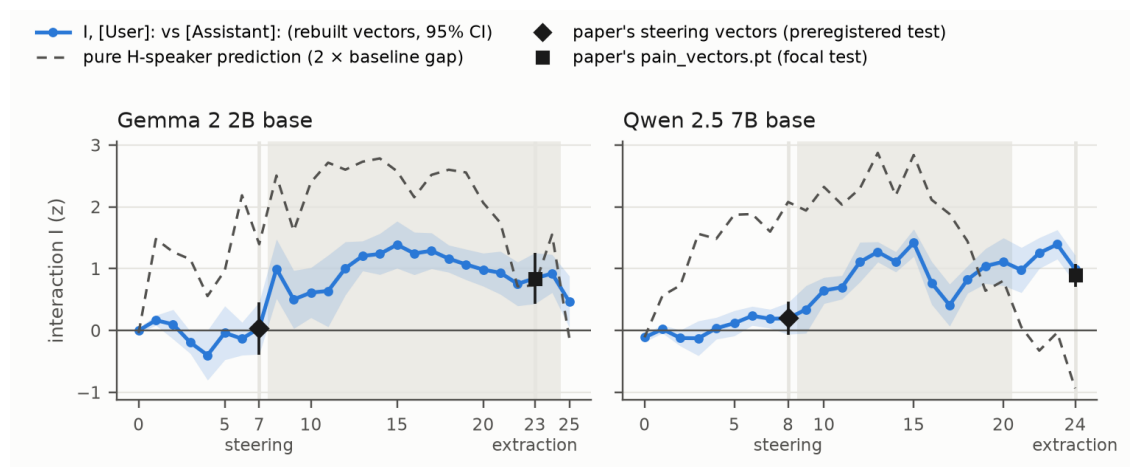


Figure 1. Interaction  $I$  by layer (positive = the suffering of the user rises relative to harm to the Assistant when the user is cued). Before the onset,  $I$  is small and inconsistent in sign ( $|I| \leq 0.4$ ). After it,  $I$  is positive at every layer with its 95% CI above zero: 0.46–1.39 z (Gemma, layers 8–25) and 0.41–1.42 z (Qwen, layers 10–24). Grey band: window layers. The dashed line ( $2 \times$  the baseline gap) falls as the paper's asymmetry weakens.

| pain axis (z), A / U / M | Gemma L23 (focal)            | Gemma L8–24 (window)         | Qwen L9–20 (window)          | Qwen L24 (focal, set aside) |
|--------------------------|------------------------------|------------------------------|------------------------------|-----------------------------|
| harm to the Assistant    | +0.10 / +0.52 / +0.57        | +0.48 / +0.43 / <b>+0.20</b> | +0.49 / +1.05 / <b>+0.09</b> | +0.15 / +0.73 / -0.42       |
| neutral                  | +0.06 / +0.70 / +0.51        | -0.45 / +0.13 / -0.38        | -0.58 / -0.22 / -0.84        | -0.89 / -0.27 / -0.88       |
| suffering of the user    | -0.27 / <b>+0.99</b> / +0.51 | -0.60 / <b>+0.34</b> / -0.34 | -0.49 / <b>+0.97</b> / -0.66 | +0.55 / +2.01 / -0.14       |

Table 2. Group means with the Assistant (A), User (U) or Moderator (M) cued next. Interaction I (User vs Assistant): Gemma L23 +0.84 [+0.43, +1.26],  $p = .0002$  (focal); Gemma window +0.99 [+0.82, +1.16] and Qwen window +0.90 [+0.79, +1.00], both  $p = .0002$ . Qwen L24 +0.89 [+0.70, +1.08],  $p = .0002$ , is set aside because its baseline gap is not positive (-0.39 [-0.96, +0.17]).

- **Gemma, layer 23 (the focal test).** The suffering of the user follows the cued voice, producing a full crossover. Harm to the Assistant is barely distinguishable from neutral under any label at this depth (elevation with the Assistant cued +0.04 [-0.68, +0.77]).
- **Both models across the window: the same partial pattern.** With the Assistant cued, harm to the Assistant is elevated (Gemma +0.93, Qwen +1.07, both  $p \leq .0005$ ) and the suffering of the user is at neutral. With the user cued, the suffering of the user rises to the level of harm to the Assistant: harm minus suffering is +0.09 [-0.62, +0.79] (Gemma) and +0.08 [-0.39, +0.54] (Qwen). The asymmetry is erased, not reversed. With a third party cued, harm to the Assistant stays above neutral (Gemma +0.57 [+0.39, +0.75], Qwen +0.93 [+0.57, +1.28]), and it does so at 12 of 17 and 12 of 12 individual layers respectively. So the suffering of the user follows the voice; harm to the Assistant largely does not.
- **Tokenization.** Qwen's [Moderator]: is one token longer than the other labels, which confounds its third-party contrast. Gemma's is a single token like the others, so Gemma's window result is the unconfounded version of the finding that harm to the Assistant persists under a third party.
- **The paper's asymmetry across depth.** Baseline gap (harm – suffering, Assistant cued), with 95% CIs: Qwen +1.06 [+0.58, +1.53] at the §4.1 layer vs -0.39 [-0.96, +0.17] at the extraction layer (change -1.45 [-1.99, -0.91],  $p = .0002$ ); Gemma +0.69 [-0.23, +1.61] vs +0.37 [-0.58, +1.32] (change -0.32 [-0.80, +0.15],  $p = .12$ ). In Qwen the asymmetry is absent at the extraction layer, but not reversed.
- **Robustness.** S1 and S2 each show the speaker component, and all five suffering categories take part. In Qwen its onset coincides with the jump in the §3.3 first- vs third-person gap (layer 10). In Gemma that gap is present from layer 0, so the §3.3 result does not simply mark where speaker attribution appears.

#### 4. What this says about H-speaker vs the self-directed reading

- **At the §4.1 readout the self-directed reading holds.** The self/other ordering does not depend on who speaks next, so it is not a turn-taking artefact. Only a small speaker component remains possible. The design cannot separate harm to the system, harm to the Assistant *character*, and hostile or rejecting content aimed at an interlocutor.
- **From mid-depth, the two sides behave differently.** The suffering of the user is weighted by whether the user is cued to speak next. Harm to the Assistant stays above neutral even when a third party is cued, in both models. H-speaker therefore describes the user's side, not the Assistant's.
- **Part of the speaker-relativity may be built into the probe.** The S1 and S2 directions were defined by contrasting pain with control sentences at the final token of first-person "... I feel:" prompts, where the pain-bearer is the voice about to speak. A direction that encodes "the upcoming voice is in pain" fits that definition by construction. Near the extraction layer, then, the persona-relativity we find may tell us as much about how the probe was made as about the model. For readers on the welfare side this changes what kind of finding it is.

- **For welfare readings.** Claims of the form “the axis tracks self-directed harm” describe the steering-layer readout, not the model's pain representations in general. Absolute levels need label-matched baselines, since the role token alone moves the early-layer axis by up to 3 z.
- **Open: state attribution or content prediction?** A late-layer signal that follows the cued voice has two readings. On one, the model represents that the upcoming speaker *is* in pain (state attribution). On the other, it represents that the upcoming *text* will be about pain (content prediction). These are different hypotheses, and treating the first as nothing over and above the second is a substantive philosophical position, arguably the crux of the welfare question. Our design cannot separate them. Stimuli that pull them apart would help: a user whose pain has resolved but who is cued to speak, or a sufferer cued to talk about something else.

## 5. Limitations and next steps

The layer sweep is exploratory; only the steering-layer test was preregistered. Qwen's speaker component rests entirely on non-focal layers, and the window rule was formulated after seeing the data, though it uses a threshold fixed in advance. There are two models and 5 categories per comparison group. [User] : -next also means “the suffering user continues their own account”. The Qwen [Moderator] : contrast is confounded by tokenization (Gemma's is not). The readout is a single token position ([ ] :), so the results may not hold elsewhere in the transcript. Replications on Llama 3.1 8B and Gemma 2 9B are preregistered and pending. Informative next steps would be a content-matched control, a third-party-target control, more suffering categories (the variation between categories limits precision), and stimuli that separate state attribution from content prediction.

*Deviations from the original plan are logged: category-level inference, a corrected power estimate (the preregistered simulation mis-scaled the noise), the exploratory sweep, its GPU-loading changes and the post hoc window analysis. Code, stimuli, per-item projections, the frozen preregistration and the deviation log are available on request. Pain-axis repository commit 7c25650.*

*Methods note: the pipeline was built and run with Claude Code (Anthropic's AI coding agent). This covers reconnaissance of the paper's repository, the scripts, the preregistration draft, the simulations, the analyses and drafts of this note. Štěpán Los made or approved every design decision at each checkpoint, and the forward passes ran on a free Google Colab T4.*